

## OSFI E-23 Checklist

| <b>E-23 Section</b>                  | <b>Checklist</b>                                                                                | <b>Yes / No / Partial</b> | <b>Comments / Gap</b> |
|--------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------|-----------------------|
| <b>1.1 Organizational Enablement</b> | Are MRM roles and responsibilities clearly defined?                                             | <input type="checkbox"/>  |                       |
|                                      | Are sufficient skilled and experienced MRM resources available?                                 | <input type="checkbox"/>  |                       |
|                                      | Is model risk reported to senior management/Board?                                              | <input type="checkbox"/>  |                       |
|                                      | Are relevant teams (Risk, Business, IT, Legal, Compliance, Ethics, etc.) involved where needed? | <input type="checkbox"/>  |                       |
| <b>1.2 MRM Framework</b>             | Is there a documented enterprise-wide MRM framework?                                            | <input type="checkbox"/>  |                       |
|                                      | Is the framework aligned with model risk appetite?                                              | <input type="checkbox"/>  |                       |
|                                      | Does it cover identification, assessment, management, monitoring and reporting of model risk?   | <input type="checkbox"/>  |                       |
|                                      | Does it cover third-party models/data?                                                          | <input type="checkbox"/>  |                       |
|                                      | Is the framework periodically reviewed?                                                         | <input type="checkbox"/>  |                       |
| <b>1.3 Use of Models</b>             | Does each model have a clear business purpose?                                                  | <input type="checkbox"/>  |                       |
|                                      | Is the model suitable and reliable for its intended use?                                        | <input type="checkbox"/>  |                       |
|                                      | Are models modified/replaced/retired when no longer fit-for-purpose?                            | <input type="checkbox"/>  |                       |

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|--------------------------------------|-----------------------------------------------------------------------------|---------------------------|-----------------------|
| <b>2.1 Model Identification</b>      | Is there a process to identify all models, including vendor models?         | <input type="checkbox"/>  |                       |
|                                      | Are models assessed to determine whether they require lifecycle governance? | <input type="checkbox"/>  |                       |
|                                      | Is there a complete and current model inventory?                            | <input type="checkbox"/>  |                       |
|                                      | Are new, changed and recently retired models tracked?                       | <input type="checkbox"/>  |                       |
| <b>2.2 Model Risk Rating</b>         | Does every applicable model have a Model Risk Rating?                       | <input type="checkbox"/>  |                       |
|                                      | Does the rating consider model vulnerability and business/material impact?  | <input type="checkbox"/>  |                       |
|                                      | Are quantitative and qualitative factors considered?                        | <input type="checkbox"/>  |                       |
|                                      | Are ratings periodically reviewed and updated after material changes?       | <input type="checkbox"/>  |                       |
|                                      | Are external/vendor models rated on a standalone basis?                     | <input type="checkbox"/>  |                       |
|                                      | Is residual model risk visible and reported?                                | <input type="checkbox"/>  |                       |
| <b>2.3 Risk Management Intensity</b> | Does Model Risk Rating determine validation scope and frequency?            | <input type="checkbox"/>  |                       |
|                                      | Does it determine documentation requirements?                               | <input type="checkbox"/>  |                       |
|                                      | Does it determine approval authority?                                       | <input type="checkbox"/>  |                       |

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|------------------------------------------------|-------------------------------------------------------------------------|---------------------------|-----------------------|
| <b>3.1 Policies, Procedures &amp; Controls</b> | Does it determine monitoring scope and frequency?                       | <input type="checkbox"/>  |                       |
|                                                | Are model-use restrictions and additional controls applied when needed? | <input type="checkbox"/>  |                       |
|                                                | Are lifecycle policies and procedures documented?                       | <input type="checkbox"/>  |                       |
|                                                | Are stakeholder responsibilities and accountability clear?              | <input type="checkbox"/>  |                       |
|                                                | Are key stakeholders involved early?                                    | <input type="checkbox"/>  |                       |
|                                                | Is independence/objectivity maintained?                                 | <input type="checkbox"/>  |                       |
|                                                | Can controls accommodate different models, including AI/ML?             | <input type="checkbox"/>  |                       |
| <b>3.2 Model Data</b>                          | Is model data accurate and fit-for-use?                                 | <input type="checkbox"/>  |                       |
|                                                | Is data relevant and representative?                                    | <input type="checkbox"/>  |                       |
|                                                | Are data ethics and customer privacy considered?                        | <input type="checkbox"/>  |                       |
|                                                | Is data lineage/provenance documented?                                  | <input type="checkbox"/>  |                       |
|                                                | Is data timely?                                                         | <input type="checkbox"/>  |                       |
|                                                | Are missing values, outliers and consistency checked?                   | <input type="checkbox"/>  |                       |
|                                                | Are synthetic/proxy data controlled and documented?                     | <input type="checkbox"/>  |                       |
|                                                | Are data cleansing controls documented?                                 | <input type="checkbox"/>  |                       |

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| <b>3.3 Model Development</b>         | Is the model methodology conceptually sound?            | <input type="checkbox"/>  |                       |
|                                      | Are model assumptions and methodology documented?       | <input type="checkbox"/>  |                       |
|                                      | Are limitations and restrictions documented?            | <input type="checkbox"/>  |                       |
|                                      | Is expert judgment documented and controlled?           | <input type="checkbox"/>  |                       |
|                                      | Are development performance tests documented?           | <input type="checkbox"/>  |                       |
|                                      | Are model/algorithm selection criteria defined?         | <input type="checkbox"/>  |                       |
|                                      | Are explainability requirements defined?                | <input type="checkbox"/>  |                       |
|                                      | Are model output/use/reporting standards defined?       | <input type="checkbox"/>  |                       |
| <b>3.4 Model Review / Validation</b> | Are monitoring criteria established during development? | <input type="checkbox"/>  |                       |
|                                      | Is validation independent from model development?       | <input type="checkbox"/>  |                       |
|                                      | Does validation confirm/challenge Model Risk Rating?    | <input type="checkbox"/>  |                       |
|                                      | Does validation assess purpose and scope?               | <input type="checkbox"/>  |                       |
|                                      | Does validation assess conceptual soundness?            | <input type="checkbox"/>  |                       |
|                                      | Are assumptions, limitations and mitigants challenged?  | <input type="checkbox"/>  |                       |

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|                             | Is data quality and appropriateness independently reviewed?                                     | <input type="checkbox"/>  |                       |
|                             | Are model outcomes/performance independently assessed?                                          | <input type="checkbox"/>  |                       |
|                             | Are AI/ML methodologies, algorithms and tools reviewed where applicable?                        | <input type="checkbox"/>  |                       |
|                             | Is model explainability assessed?                                                               | <input type="checkbox"/>  |                       |
|                             | Are third-party models, data, platforms and libraries reviewed?                                 | <input type="checkbox"/>  |                       |
|                             | Are validation findings and actions documented?                                                 | <input type="checkbox"/>  |                       |
|                             | Does validation provide an overall recommendation to the Model Approver?                        | <input type="checkbox"/>  |                       |
|                             | Are new models, changes, performance breaches and significant data changes validation triggers? | <input type="checkbox"/>  |                       |
| <b>Model Approval</b>       | Is suitability for production/use formally assessed?                                            | <input type="checkbox"/>  |                       |
|                             | Is Model Risk Rating affirmed during approval?                                                  | <input type="checkbox"/>  |                       |
|                             | Is residual risk considered?                                                                    | <input type="checkbox"/>  |                       |
|                             | Are compensating controls required for accepted weaknesses?                                     | <input type="checkbox"/>  |                       |
| <b>3.5 Model Deployment</b> | Is the approved model correctly implemented in production?                                      | <input type="checkbox"/>  |                       |

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|-------------------------------|----------------------------------------------------------------|---------------------------|-----------------------|
|                               | Is development data consistent with production data?           | <input type="checkbox"/>  |                       |
|                               | Is production testing performed?                               | <input type="checkbox"/>  |                       |
|                               | Are deployment roles and approvals documented?                 | <input type="checkbox"/>  |                       |
|                               | Are change controls established?                               | <input type="checkbox"/>  |                       |
|                               | Are overlays/exceptions controlled?                            | <input type="checkbox"/>  |                       |
|                               | Are cybersecurity/infrastructure risks considered?             | <input type="checkbox"/>  |                       |
| <b>3.6 Model Monitoring</b>   | Are monitoring frequency and scope defined based on risk?      | <input type="checkbox"/>  |                       |
|                               | Are quantitative and qualitative monitoring measures defined?  | <input type="checkbox"/>  |                       |
|                               | Are model performance, usage, data and dependencies monitored? | <input type="checkbox"/>  |                       |
|                               | Are performance thresholds/breaches defined?                   | <input type="checkbox"/>  |                       |
|                               | Are material-change criteria defined?                          | <input type="checkbox"/>  |                       |
|                               | Are contingency and escalation procedures established?         | <input type="checkbox"/>  |                       |
|                               | Is model drift monitored, particularly for AI/ML?              | <input type="checkbox"/>  |                       |
| <b>3.6 Model Decommission</b> | Are criteria/processes established for retiring models?        | <input type="checkbox"/>  |                       |
|                               | Are stakeholders notified?                                     | <input type="checkbox"/>  |                       |

| E-23 Section                      | Checklist                                                    | Yes / No<br>/ Partial<br>Gap | Comments / |
|-----------------------------------|--------------------------------------------------------------|------------------------------|------------|
|                                   | Are retired models and documentation retained appropriately? | <input type="checkbox"/>     |            |
|                                   | Are third-party model retirement requirements addressed?     | <input type="checkbox"/>     |            |
|                                   | Are downstream impacts monitored?                            | <input type="checkbox"/>     |            |
| <b>Appendix – Model Inventory</b> | Does the inventory contain model ID/name/description?        | <input type="checkbox"/>     |            |
|                                   | Risk rating, owner, developer and origin?                    | <input type="checkbox"/>     |            |
|                                   | Version and production date?                                 | <input type="checkbox"/>     |            |
|                                   | Reviewer and approver?                                       | <input type="checkbox"/>     |            |
|                                   | Dependencies and data sources?                               | <input type="checkbox"/>     |            |
|                                   | Approved uses and limitations?                               | <input type="checkbox"/>     |            |
|                                   | Last review date, monitoring status and next review date?    | <input type="checkbox"/>     |            |